

Advanced Materials and Condensed Matter Physics

Prof. Aftab Alam leads groundbreaking research in magnetism, spintronics, and exotic phases of superconductors and topological insulators. His current work focuses on mixed metal-antimony oxide nanocomposites for low-pH water oxidation electrocatalysts with outstanding durability at ambient and elevated temperatures, antiferro-quadrupolar ordering in cerium hexaboride CeB_6 , double perovskites as highly stable and efficient solar harvesting materials, and lead-free magnetic double perovskites for photovoltaic and photocatalysis applications. His group extensively studies thermoelectric properties of materials with variable relaxation time effects, bismuth-based half-Heusler alloys with giant thermoelectric figures of merit, and electronic structure and thermodynamics of transition metal alloys including vibrational properties and response functions in materials with defects.^{[1][2][3]}

Prof. M. Aslam specializes in quantum/nano material fabrication with focus on oxides and earth-abundant compounds (CZTS) for photovoltaic applications, graphene and graphene-like semiconductors for water splitting, photosensing and energy storage applications. His recent work includes electrochemical synthesis of graphene quantum dots from graphene oxide for soil moisture sensing, defect-related emissions in ZnO nanorods, iron oxide nanorods as high-performance MRI contrast agents, and surface-controlled synthesis of MFe_2O_4 nanoparticles. His group works on organic perovskites for nanoparticle fabrication and stabilization, magnetic nanoparticles for MRI imaging and GMR applications, and highly sensitive MoS_2 nanosheets as stable humidity sensors.^{[4][5][6][7]}

Prof. Sayantika Bhowal conducts cutting-edge theoretical research on hidden charge and magnetic order, charge/spin/orbital Hall effects, and magnetic skyrmions. Her recent publications include work on electric toroidal dipole order and hidden spin polarization in ferroaxial materials, surface magnetization in antiferromagnets and its relation to magnetoelectric responses, ferroically ordered magnetic octupoles in d-wave altermagnets, and polar metals principles and prospects. She studies magnetism and spin-orbitronics, magnetoelectrics, topology, ferroelectrics, Compton scattering applications for revealing hidden magnetoelectric multipoles, and orbital Hall effects in transition metal dichalcogenides.^{[8][9][10]}

Prof. Subhabrata Dhar focuses on growth, transport and optical properties of semiconductors, particularly low-dimensional semiconductors, semiconductor nanostructures, and magnetic semiconductors. His research includes high electron mobility through edge states in random networks of c-axis oriented wedge-shaped GaN nanowalls, colossal magnetic moment phenomena in Gd-doped GaN, origin of high-temperature ferromagnetism in (Ga,Mn)N layers, and optical detection of defects

associated with magnetic properties in GaN:Gd layers. His group works on molecular beam epitaxy growth techniques and characterization of wide bandgap semiconductors.^{[11][12]}

Quantum Information, Computing and Optics

Prof. Himadri Shekhar Dhar leads theoretical research in quantum information, entanglement and resource theories, quantum optics, and cavity QED systems. His recent work includes Bose-Einstein condensation of light in semiconductor quantum well microcavities, breakdown of temporal coherence in photon condensates, periodic cavity state revivals from atomic frequency combs, and sensing atomic superfluid rotation beyond the standard quantum limit. His research covers quantum information dynamics using tensor network methods, quantum trajectories and variational quantum algorithms, optimal control theory for quantum device design, and machine learning applications in quantum systems.^{[13][14][15]}

Prof. Siddhasatta Mahapatra specializes in quantum electronics, quantum information, and quantum computation with focus on silicon-based quantum devices. His groundbreaking work includes single-atom transistors, spin blockade and exchange in Coulomb-confined silicon double quantum dots, and charge sensing of precisely positioned phosphorus donors in silicon. Recent projects involve arbitrary waveform generation for Si-quantum dot qubit control, development of silicon/germanium spintronics, and thermoelectric materials and modules. His group leads the Centre of Excellence in Quantum Information, Communication Science and Technology (CoE-QuICST) at IIT Bombay.^{[16][17][18][19]}

Prof. Uditendu Mukhopadhyay focuses on experimental quantum simulation and quantum computation using gate-defined quantum dot arrays. His research involves simulating the Fermi-Hubbard model using two-dimensional arrays of quantum dots, creating 2x2 quantum dot arrays with unprecedented tunability of tunnel coupling, and achieving individual control over chemical potential and electron occupation. His work demonstrates Nagaoka ferromagnetism in quantum dot systems and explores artificial lattices of interacting particles.^{[20][21][22]}

Astrophysics and Gravitational Physics

Prof. Varun Bhalerao leads astrophysics research focusing on electromagnetic counterparts of gravitational wave sources, X-ray telescopes development, and neutron star studies. His group operates GROWTH-India, India's first fully robotic telescope, and leads Indian efforts in studying electromagnetic counterparts to gravitational wave sources. Recent work includes hunting for light from binary black hole and neutron star mergers using AstroSat's CZTI instrument, and development of the proposed Daksha

mission for high-energy transient detection. His research covers multi-messenger astronomy, kilonova observations, and X-ray studies of transient sources.^{[23][24][25][26]}

Prof. Rahul Kashyap specializes in computational astrophysics and multimessenger signals from binary compact mergers. His research involves hydrodynamical simulations of binary white dwarf and binary neutron star mergers, kilonova ejecta modeling, and astrophysical implications from parametric Bayesian inferences using gravitational wave sources. He works on Type Ia supernovae from binary white dwarf mergers, one-armed spiral instability in post-merger accretion disks, and standardization of kilonova light curves. As a LIGO-Virgo collaboration member, he contributes to gravitational wave data analysis and cosmic Explorer design.^{[27][28][29][30]}

Particle Physics and High Energy Theory

Prof. Asmita Mukherjee conducts theoretical particle physics research in QCD, QCD spin physics, light-cone wave functions, and collider phenomenology. Her work focuses on probing matter structure in three dimensions through deep inelastic scattering, preparation for the upcoming Electron-Ion Collider (EIC) at Brookhaven National Lab, and understanding nucleon tomography. She studies parton distributions, generalized parton distributions, and transverse momentum dependent distributions, contributing to understanding how quarks and gluons bind together to form protons and neutrons.^{[31][32][33]}

Active Matter and Biological Physics

Prof. Nitin Kumar leads experimental research in active matter, living matter, soft condensed matter, and biological physics. His group develops programmable robotic active matter systems to study collective behavior, flocking phenomena, and emergence of run-and-tumble dynamics in coupled self-propelled robots. Recent work includes tunable structure and dynamics of active liquid crystals, spatiotemporal control through activity patterning, and trapping and sorting of active particles through motility-induced condensation. His research covers active granular matter, liquid crystal physics, and statistical physics of living systems.^{[34][35][36][37]}

Prof. Raghunath Chelakkot specializes in driven soft matter, active matter, and physics of sentient systems. His research includes percolation transitions in binary mixtures of active Brownian particles, morphodynamics of invasion by active rods, synchronized oscillations and jammed clusters in active filament arrays, and mechano-sensing mechanisms in plant roots. He studies topological dynamics of rapid non-planar gaits in slithering snakes, flagellar dynamics of connected chains of active particles, and flow-induced helical coiling of semiflexible polymers in microchannels.^{[38][39][40]}

Prof. Mithun Kumar Mitra focuses on spatial organization of the 3D genome, epigenetic landscapes, early embryonic development, and intracellular transport by molecular motors. His research includes lamin-associated chromatin models for chromosome organization, the role of nucleosomal barriers in cohesin-mediated loop formation, delayed self-regulation in epigenetic landscapes, and catch bonding effects in dynein-mediated cellular transport. Recent work covers protein search processes mediated by chromatin topology, liquid-liquid phase separation of lamin in cardiomyopathic mutations, and theoretical analysis of cargo transport by catch bonded motors.^{[41][42][43]}

Many-Body Physics and Quantum Criticality

Prof. Soumya Bera conducts theoretical research in many-body localization, non-equilibrium dynamics, disorder-driven quantum phase transitions, and Floquet systems. His work includes the internal clock of many-body delocalization, distinguishing dynamical quantum criticality through local fidelity distances, quantum Hall criticality in amorphous Chern insulators, and finite-size prethermalization at chaos-to-integrable crossover. His research covers open quantum systems, topological materials, subdiffusion in Anderson models, and spin dynamics in Heisenberg chains.^{[44][45][46][47]}

Prof. Prashant Kumar specializes in topological phases and phase transitions, quantum Hall physics, and strongly correlated systems. His research focuses on dual perspectives on quantum Hall plateau transitions, bounds in correlations of fractional quantum Hall states, scaling of entanglement entropy at quantum critical points, and interaction effects on quantum Hall transitions including dynamical scaling laws and superuniversality. He uses density-matrix renormalization group methods and exact diagonalization to study neutral excitations of quantum Hall states.^{[48][49][50][51][52]}

Nuclear Physics and Spectroscopy

Prof. Pragya Das conducts experimental nuclear structure studies using heavy-ion fusion reactions and simulation-based imaging with scattered photons in PET. Her research includes nuclei excited to high-spin states using the 14 MV Pelletron accelerator, spectroscopic properties investigation with the Indian National Gamma Array (INGA), and measurement of short lifetimes using DSAM technique. She studies nuclear chirality, shape coexistence, signature aspects, and different nuclear shapes (prolate, oblate, triaxial). Her simulation work focuses on improving PET imaging by utilizing tissue-scattered data for enhanced statistical image quality.^[53]

Prof. Dibyendu Das specializes in theory of stochastic processes and non-equilibrium statistical physics with applications to biological systems. His research includes exact distribution of quantal content in

synaptic transmission, first passage problems in gene expression, protein threshold crossing dynamics, and stochastic resetting in diffusive search processes. Recent work covers synaptic transmission distribution analysis, competitive advantage of stochastic resetting over potential bias in target search, and experimental studies of first passage of active floaters.^{[54][55][56][57]}

Attosecond Physics and Ultrafast Science

Prof. Gopal Dixit leads theoretical attosecond physics research focusing on ultrafast electron dynamics in atoms, molecules, and condensed matter. His work contributes to the Nobel Prize-winning field of attosecond physics, including light-induced valleytronics in pristine graphene, high-harmonic spectroscopy of light-driven anomalous Hall effects in Weyl semimetals, and tailoring photocurrent in Weyl semimetals via intense laser irradiation. His research covers all-optical ultrafast valley switching in 2D materials, generation of circularly-polarized high-harmonics, and four-dimensional imaging of lattice dynamics using ab-initio simulations.^{[58][59][60]}

Nanophotonics and 2D Materials

Prof. Anshuman Kumar conducts experimental and theoretical research in nanophotonics and 2D materials. His work includes polaritons in layered 2D materials, tunable light-matter interaction in graphene-hBN systems, planar hyperbolic polaritons in van der Waals materials, and gate-tunable light-matter interaction in natural biaxial hyperbolic heterostructures. Recent research covers mid-infrared polarization engineering via sub-wavelength biaxial hyperbolic crystals, fast design of plasmonic metasurfaces enabled by deep learning, and boosting quantum yields in 2D semiconductors via proximal metal plates.^{[61][62][63][64]}

Emerging Materials and Device Physics

Prof. Dinesh Kabra specializes in emerging photonic materials including halide perovskites and molecular semiconductors. His research focuses on micro-spectroscopy techniques (emission, transient absorption, Raman), physics and engineering of optoelectronic devices including solar cells, LEDs and detectors. Recent work includes optical and optoelectronic properties of 2D, quasi-2D and 3D metal halide perovskites, NIR-transparent organic and perovskite bi-facial solar cells, and contrasting temperature dependence of band gaps in halide perovskites. He co-founded ART-PV India Pvt Ltd to commercialize perovskite solar cell technology.^{[65][66][67][68]}

Prof. Pramod Kumar conducts research on nanostructure-based sensors, organic/inorganic field effect transistors, and piezoelectric devices. His work includes seedless hydrothermal growth of ZnO nanorods for flexible tactile sensors, charge transport mechanisms in organic semiconductors, and integration of artificial intelligence in sensor analysis. He studies high band-gap semiconductors, organic semiconductors, energy generation and storage devices, and the effect of molecular arrangement on charge carrier transport.^{[69][70][71][72]}

Advanced Spectroscopy and Magnetic Materials

Prof. Avinash V. Mahajan leads research in novel magnetic states, low-dimensional systems, and quantum spin liquids. His work includes NMR/NQR studies of frustrated magnetic systems, investigation of spin-1/2 cooperative paramagnets in diluted triangular lattices, gapless quantum spin liquids in triangular systems, and signatures of spin-orbit coupling in honeycomb materials. His group operates customized NMR/NQR systems for low-temperature experiments on powder and single-crystal samples, studying magnetic properties of various exotic compounds.^{[73][74][75][76]}

Prof. Tapanendu Kundu focuses on photonics and optoelectronics development for optical sensing applications, laser spectroscopy, and plasmonics. His research includes photoresponse of CVD-grown crystalline quantum dot-embedded covalent organic framework thin films, detection of plant viruses using attenuated total reflection-mediated localized surface-plasmon-resonance platforms, and spatio-temporal analysis of electric field-induced solid-state reduction dynamics of graphene oxide thin films. His work covers light-matter interaction phenomena, optical MEMS, and laser fundamentals.^{[77][78][79]}

Mesoscopic Physics and Quantum Transport

Prof. Kantimay Das Gupta specializes in mesoscopic physics, low-temperature transport, semiconductor heterostructures, and magnetic levitation. His research includes experimental evidence of superconducting layers in epitaxial indium nitride, switching between attractive and repulsive Coulomb interaction mediated drag in ambipolar bilayer devices, and fabrication of gallium arsenide ambipolar quantum point contacts. His work covers effect of parameter mismatch on synchronization of coupled self-sustained oscillators and quantum Hall effects in various systems.^{[80][81][82][83]}